

Layer Architecture — architecture-model-standard

Layer Overview

The system uses a single **Application Layer** (L-APP) containing all 9 top-level components. Two of these — COMP-CORE and COMP-MANIFEST — are decomposed into sub-components, bringing the total to 24 addressable units.

Layer	ID	Components	Sub-Components	Total
Application	L-APP	9	15	24

Component Catalog

Top-Level Components

ID	Name	Description
COMP-CORE	core	Parser, validator, slicer, differ, merger, decomposer, type system
COMP-MANIFEST	manifest	AST scanner, block discovery, metrics, interfaces, body hints, test analyzer
COMP-CLI	cli	CLI entry point (9 commands)
COMP-CONFIG	config	Configuration loader and schema
COMP-SPEC	spec	JSON Schema definitions
COMP-EXTRACT	extract	Model extraction from source code
COMP-PROFILES	profiles	Domain profile loading, open enums
COMP-UTILS	utils	Shared file discovery, exclusion patterns
COMP-ENRICH	enrich	Auto-enrichment of models from AST

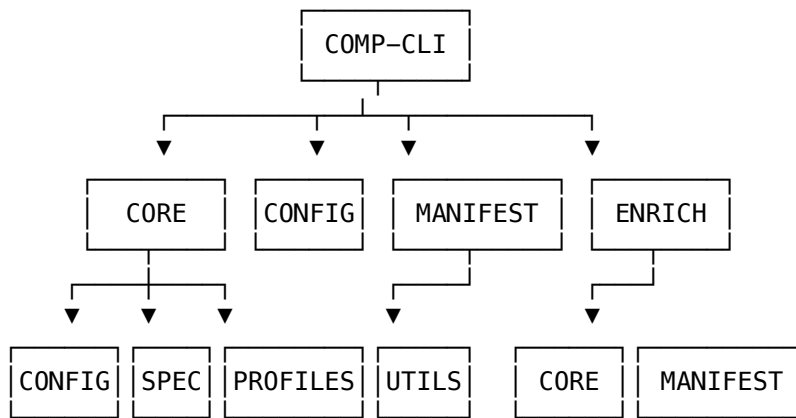
COMP-CORE Sub-Components (7)

ID	Name	Role
COMP-CORE-PARSER	parser	Load/save architecture models from YAML
COMP-CORE-VALIDATOR	validator	Structural validation, scoring
COMP-CORE-SLICER	slicer	Extract sub-models by F-block or layer
COMP-CORE-DIFFER	differ	Diff two architecture models
COMP-CORE-MERGER	merger	Merge partial models
COMP-CORE-DECOMPOSER	decomposer	Decompose monolith into subsystems
COMP-CORE-TYPES	types	Shared type definitions (ArchitectureModel, etc.)

COMP-MANIFEST Sub-Components (8)

ID	Name	Role
COMP-MANIFEST-GENERATOR	generator	Orchestrates manifest generation
COMP-MANIFEST-SCANNER	scanner	AST scanning of Python source files
COMP-MANIFEST-BLOCKS	blocks	F-block discovery from package structure
COMP-MANIFEST-METRICS	metrics	LOC, complexity, function counts
COMP-MANIFEST-INTERFACES	interfaces	Public API extraction
COMP-MANIFEST-BODY-HINTS	body-hints	Trivial function body extraction
COMP-MANIFEST-TEST-ANALYZER	test-analyzer	Test file analysis, contract extraction
COMP-MANIFEST-TYPES	types	Manifest-specific type definitions

Top-Level Dependency Graph



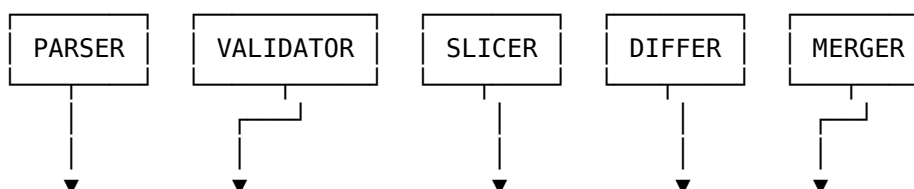
COMP-EXTRACT → COMP-CORE

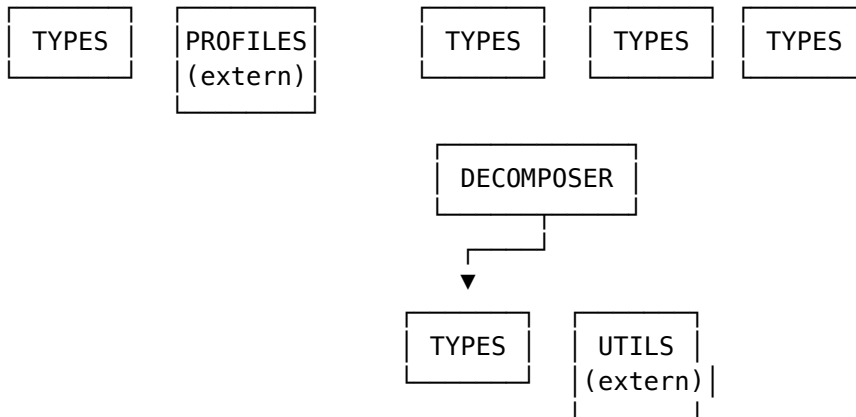
Dependency summary:

- **COMP-CLI** □ CORE, CONFIG, MANIFEST, ENRICH
- **COMP-CORE** □ CONFIG, SPEC, PROFILES
- **COMP-MANIFEST** □ CONFIG, UTILS
- **COMP-EXTRACT** □ CORE
- **COMP-ENRICH** □ CORE, MANIFEST

Leaf components (no outgoing dependencies): CONFIG, SPEC, PROFILES, UTILS.

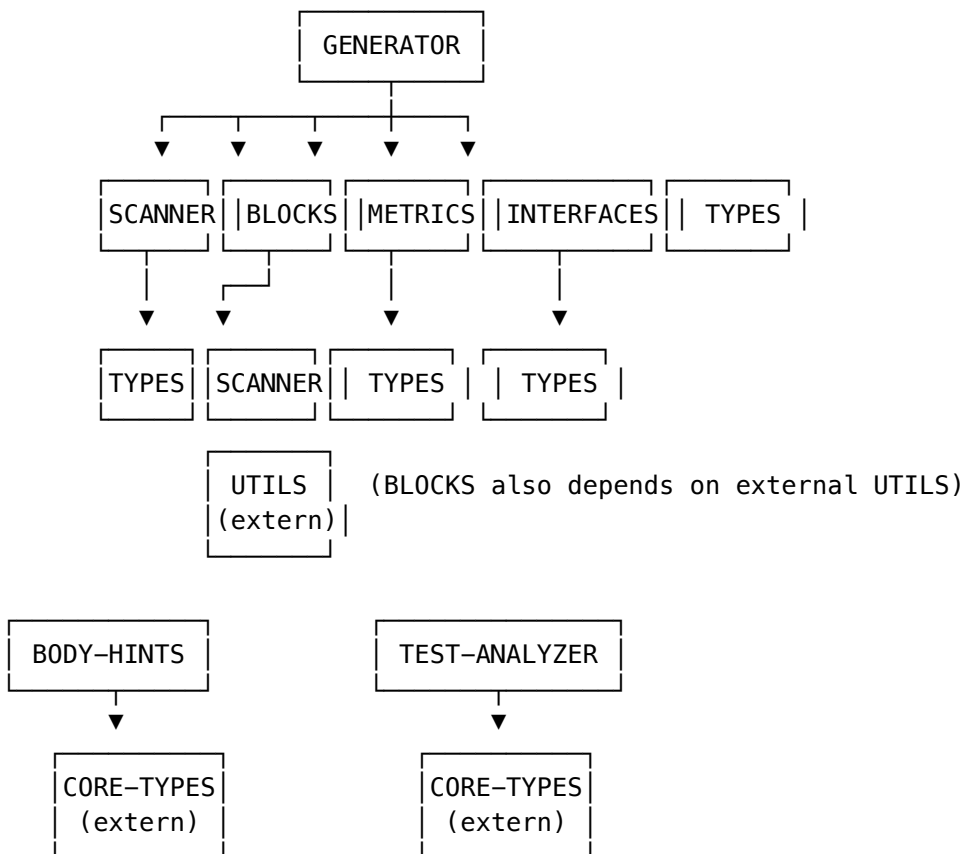
COMP-CORE Internal Dependency Graph





All six operational sub-components depend on **TYPES**. VALIDATOR also reaches outside to PROFILES. MERGER and DECOMPOSER reach outside to UTILS.

COMP-MANIFEST Internal Dependency Graph



GENERATOR orchestrates SCANNER, BLOCKS, METRICS, INTERFACES, and TYPES. BODY-HINTS and TEST-ANALYZER depend on CORE-TYPES (cross-component).

Interface Exposure Mapping

Component	Interface ID	API Surface
COMP-CLI	IF-CLI	9 CLI commands (init, extract, validate, slice, diff, query, context, stats, impact)
COMP-CORE-PARSER	IF-PARSE-API	load_model(), save_model()
COMP-CORE-VALIDATOR	IF-VALIDATE-API	validate_model() → ValidationResult
COMP-CORE-SLICER	IF-SLICER-API	slice_by_fblock(), slice_by_layer()
COMP-MANIFEST-GENERATOR	IF-MANIFEST-API	generate_manifest() → Manifest
COMP-PROFILES	IF-PROFILE-API	load_profile() → DomainProfile
COMP-ENRICH	IF-ENRICH-API	enrich_model() → ArchitectureModel

Dependency Matrix

Rows depend on columns. • = direct dependency.

	CONFIG	SPEC	PROFILES	UTILS	CORE	MANIFEST
COMP-CLI	•				•	•
COMP-CORE	•	•	•			
COMP-MANIFEST	•			•		
COMP-EXTRACT					•	
COMP-ENRICH					•	•

COMP-CLI also depends on COMP-ENRICH (omitted from columns for space — it is a transitive consumer of CORE and MANIFEST through ENRICH).

Technology Stack

Technology	Role
Python 3.10+	Implementation language
YAML	Model serialization format
JSON Schema	Model structural validation (COMP-SPEC)
Python ast module	Source code scanning (COMP-MANIFEST)
Click	CLI framework (COMP-CLI)

Relationship Summary

Type	Count
depends-on	34
contains	24
realizes	10
traces-to	9
exposes	7

Type	Count
consumes	4
constrained-by	3
Total	91